Jagraj Gill

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EDUCATION

Simon Fraser University, **December 2027**
Bachelor of Science in Computing Science
Relevant Coursework: Data Structures & Algorithms I, Data Structures & Algorithms II, Software Engineering, Databases

EXPERIENCE

Software Engineer Intern, Savi Finance **– Toronto, ON** **May 2025 – December 2025**

- **Developed** an **AI-powered task scheduling** system that ranked tasks by priority and immediacy, reducing manual planning time by **10%**
- Engineered shared modules across **React** and **React Native** backed by **MongoDB** services for a fintech SaaS platform (**1000+ users**), reducing release cycles by **15%** and improving code reuse across web and mobile.
- Built and integrated **image upload workflow** using **AWS S3** and **GraphQL**, supporting custom image previews and improving upload reliability while reducing image load failures by **10%**
- Refactored **login UI and form validation** using **React** and custom hooks, adding real-time feedback and reducing user login errors by **45%,** improving authentication success rates and onboarding efficiency

Hackathon, Storm Hacks 2024 **– Burnaby, BC** **May 2024**

- **Directed a 2-person** team in a 24-hour hackathon, coordinating task division and project delivery under strict time constraints
- Designed and deployed **REST APIs** with **Node.js** and **Express.js** to enable real-time data updates, **reducing system response time by 30%**

PROJECTS

AI Study Pal | Python, FastAPI, Gemini API, HTML, CSS, JavaScript **May 2026**

- Built an **end-to-end pipeline** that converts PDF documents into structured flashcard datasets by extracting, cleaning, and processing unstructured text with **LLM APIs**
- Designed and enforced a **strict JSON schema** for AI generated flashcards with question answer pairs, improving output consistency and enabling reliable downstream parsing
- Implemented **validation logic in Python** to verify data structure, field integrity, and content quality, preventing malformed or incomplete AI responses from entering the system
- Developed an **interactive web interface** using HTML, CSS, and JavaScript to display flashcards with flip and navigation features, enabling real time study from generated content

Wireless 4-DOF Robotic Arm | ESP32, C++, Embedded Systems **February 2026**

- Designed and fabricated a **4-DOF robotic arm** using **ESP32 microcontrollers, MG90S servos,** and custom **3D printed components**
- Developed **embedded C++ control software** to process analog sensor input and generate **PWM signals** for real-time actuator control
- Implemented **wireless control between dual ESP32 devices**, enabling synchronized motion through real-time data transmission
- Engineered **modular control architecture** to support future expansion including **inverse kinematics** and advanced sensor integration

Investment Tracker | Java, Swing, JFreeChart **May 2025**

- Built a **Java desktop app** with a **Swing based GUI** to visualize investment growth, **rendering side by side charts** using **JFreeChart**
- Engineered portfolio logic to calculate **asset performance in real time**, based on live spot prices, quantities, and historical purchase data
- Improved user reliability through **exception handling** and **input validation**, ensuring consistent chart behavior for dynamic asset entries

SKILLS

Languages: C, C++, Embedded C, Python, Java, JavaScript, x86 Assembly
Web & Backend: React, HTML5, CSS, Node.js, Express.js, GraphQL
Databases & Tools: MongoDB, MySQL, SQLite, AWS S3, Git, Linux
Embedded Systems: Microcontrollers, RTOS, Hardware Interfacing, I2C, UART